



$$\vec{r} = \vec{r}_2 - \vec{r}_1 \quad V = V(r) \neq V(\vec{r}_1, \vec{r}_2)$$

$$\vec{r}_r = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} \quad \hookrightarrow \vec{F} = F \frac{\vec{r}}{r}, \quad \vec{F} = -\nabla V \rightarrow V = V(r) \neq V(\vec{r}) \Rightarrow \vec{F} = -\frac{dV}{dr} \frac{\vec{r}}{r}$$

$$L = \frac{1}{2} m_1 \dot{r}_1^2 + \frac{1}{2} m_2 \dot{r}_2^2 + V(r) \rightarrow L = \underbrace{\frac{1}{2} (m_1 + m_2) \dot{\vec{r}}_r^2}_{\text{konst.}} + \underbrace{\frac{1}{2} \mu \dot{r}^2 + V(r)}_{L'}$$

$$\vec{M} = \vec{r} \times \vec{F} = 0 \Rightarrow \vec{L} = \text{konst.}$$

$$\vec{L} = \vec{r} \times \vec{p} = \text{konst.} \Rightarrow \vec{r}, \vec{p} \text{ ostajata v isti ravlini}$$

① Delec v potencialu $-\frac{k}{r}$

$$r(\varphi) = ?$$

$$L = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\varphi}^2) + \frac{k}{r}$$

$$m r^2 \ddot{\varphi} = p_{\varphi} \quad \frac{d}{dt} [m r \dot{\varphi}] + \frac{k}{r^2} = 0$$

$$\frac{d\varphi}{dt} \frac{d}{d\varphi} \left[m \underbrace{\dot{\varphi} \frac{dr}{d\varphi}}_{r'} \right] + \frac{k}{r^2} = 0$$

$$\dot{\varphi} \frac{d}{d\varphi} \left(\frac{p_{\varphi}}{r^2} r' \right) - \frac{p_{\varphi}^2}{m r^3} + \frac{k}{r^2} = 0$$

$$\frac{p_{\varphi}^2}{m} \frac{d^2}{d\varphi^2} \left(-\frac{1}{r} \right) - \frac{p_{\varphi}^2}{m r^3} + \frac{k}{r^2} = 0 \quad u = \frac{1}{r}$$

$$\frac{p_{\varphi}^2}{m} u'' + \frac{p_{\varphi}^2}{m} u = k$$

$$\left. \begin{array}{l} h: u = A \cos(\varphi - \varphi_0) \\ p: u = \frac{k m}{p_{\varphi}^2} \end{array} \right\} u(\varphi) = A \cos(\varphi - \varphi_0) + \frac{k m}{p_{\varphi}^2}$$

$$r(\varphi) = \frac{1}{A \cos(\varphi - \varphi_0) + \frac{k m}{p_{\varphi}^2}}$$

$$r' = \frac{A \sin(\varphi - \varphi_0)}{\frac{k m}{p_{\varphi}^2} + A \cos(\varphi - \varphi_0)} \rightarrow r_0: r'(\varphi_0) = 0 \rightarrow r_0 = r(\varphi_0) = \frac{1}{A + \frac{k m}{p_{\varphi}^2}}$$

$$H = T + V = \dots = \frac{p_\varphi^2}{2\mu r_0^2} - \frac{k}{r_0} = \frac{p_\varphi^2}{2\mu} \underbrace{\left(A + \frac{k\mu}{p_\varphi^2}\right)^2}_B - k\left(A + \frac{k\mu}{p_\varphi^2}\right)$$

$$\frac{p_\varphi^2}{2\mu} B^2 - kB - H = 0$$

$$A + \frac{k\mu}{p_\varphi^2} = \frac{k \pm \sqrt{k^2 + 4H \frac{p_\varphi^2}{2\mu}}}{\frac{p_\varphi^2}{2\mu}} \rightarrow A = \frac{\pm \sqrt{k^2 + 4H \frac{p_\varphi^2}{2\mu}}}{p_\varphi^2} 2\mu$$

$$\rightarrow u(\varphi) = A \cos(\varphi - \varphi_0) + \frac{k\mu}{p_\varphi^2} = \frac{\pm 2\mu \sqrt{k^2 + 4H \frac{p_\varphi^2}{2\mu}} \cos(\varphi - \varphi_0) + k\mu}{p_\varphi^2}$$

$$r(\varphi) = \frac{p_\varphi^2}{2\mu} \frac{1}{1 \pm \sqrt{1 + \frac{4H p_\varphi^2}{2k^2 \mu} \cos(\varphi - \varphi_0)}} = \frac{p}{1 + \varepsilon \cos(\varphi - \varphi_0)}$$

↓

$\varepsilon = 0$: $r = \text{konst.} \rightarrow$ krožnica

$|\varepsilon| = 1$: parabola

$|\varepsilon| > 1$: hiperbola

$0 < |\varepsilon| < 1$: $H < 0$ elipsa

$$V_{\text{eff}} = \frac{p_\varphi^2}{2\mu r^2} - \frac{k}{r}$$

